Silver Tsunami: One-Page Execution Blueprint

The Silver Tsunami: Unearthing the Demographic Forces Shaping Global Population Ageing

Goal: group countries into age stages in 2000, see who stayed and who moved up by 2020 (and by 2040, projected), and find what explains the difference.

Setup

Item	Choice
Data	UN WPP 2024 master file, plus the UN age-structure file (65+ share)
Countries	161 with 1M+ people (`TPopulation1July >= 1000`)
Base year / checks	2000 / 2020, then 2040 as a projection only
Clustering	k-means, k = 3, on `MedianAgePop` only: Young, Transitioning, Older

Execution steps

Phase	Task	Output
1. Prepare	Load the file in RStudio, filter the 161 countries, merge the age-structure file on `ISO3_code` and `Time`	Clean dataset
2. Cluster	`kmeans(MedianAgePop, 3, nstart = 25)` on 2000. Sort centres. Check silhouette for k = 2 to 5	Stage labels, validity plot
3. Track	Assign 2020 and 2040 to the nearest 2000 centre. Build transition tables	Stayers and movers counts

Phase	Task	Output
4. Drivers	Regress the rise in median age (2000 to 2020) on lagged fertility, change in life expectancy at 65, change in adult mortality, net migration, age at childbearing, starting age	Effect sizes with bootstrap intervals
5. Diagnose	Residual plots, variance inflation factors, influence (Cook's distance), robust or ridge version	Diagnostics table
6. Robustness	Repeat with k = 2 and 4, base year 1990 or 2010, TFR added to clustering	Robustness table
7. Visuals	Three-map row (2000, 2020, 2040), alluvial chart, driver effects plot	3 figures
8. Deliver	Build 4 slides, write the script, record and time the video	Deck and video

Rules to stay safe

- Keep birth rate, death rate, natural change and growth rate **out** of the driver model: they follow from the age structure.
- Never test drivers on 2025+ values, because they are projections built on assumptions.
- Say "associated with", not "caused by".
- Expect collinearity between past fertility and starting age; report it and use ridge as a check.

The 4 slides (5-minute video)

#	Message	Visual	Time
1	Countries sit in three age stages	Cluster plot	60s
2	Movement is one-way: 50 moved up, none moved down (2000 to 2020)	Maps and transition table	90s

#	Message	Visual	Time
3	Past fertility is the main driver; longevity second; migration little	Driver effects plot	90s
4	By 2040 most countries are Transitioning or Older; a small group in Africa stays young	2040 map, recommendations	60s

Deliverables checklist

- Clean R script and a short repo README
- Transition table, diagnostics table and robustness table
- 4 slides and a 5-minute video, rehearsed against a timer

Biggest risks

- **Slide overload:** one message per slide.
- **Weak diagnostics:** show them, even briefly.
- **Overclaiming:** state limits on slide 4.
- **Rules:** confirm the competition's rules on external data before using the age-structure file.